Assignment 6

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In this example we will compare the accuracies of exact discretization and approximate discretization. To perform this, consider the system in Problem 3(a) with $U(t) = 0$ and carry out the following steps:

(i) Plot t vs. X(t) (both components of X(t)) for $0 \leq t \leq 2$.

(ii) Pick $h = 0.5$ and consider the approximate discretization of the system. Plot t versus $X(t)$ on the same figure as the one obtained from part (i) [you need to use the 'hold on' command in Matlab]. Use the Matlab code:

```
h= 0.5; t = 0: h : 2;

hold on; plot(t, X);
```

(iii) Pick h = 0.5 and consider the exact discretization of the system. Plot t vs $X(t)$ on the same figure.

(iv) Carry out parts (ii)–(iii) by choosing $h = 0.005$.

(v) Comment on the quality of the approximations from these plots. Make sure to comment on the two following quantities:

(1) For a fixed h, which approximation is closer is the solution of the continuous time system, and

(2) As h decreases, do the two discretization methods (exact and approximate) become indistinguishable?

System from problem 3(a):

$$\dot{X}(t) = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix} X(t) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} U(t)$$

Where $U(t)$ is set to zero.

```
clc;clear;close all;

% assume initial conditions since not provided
x0 = [1;1];
A = [0 1; 1 -1]
```

```
A = 2x2
     0     1
     1    -1
```

```
%[V,D] = eig(A);
t = linspace(0,2,100);
expm(A)
```

```
ans = 2x2
    1.3973    0.7410
    0.7410    0.6563
```

```
%V*expm(D)*V';
X1 =@(t) expm(A.*t)
```

```
X1 = function_handle with value:
    @(t)expm(A.*t)
```

```
X1(2)*x0
```

```
ans = 2x1
    4.0233
    2.5016
```

```
% for loop to populate X(t)
X = zeros(2, numel(t));
for k = 1:numel(t)
    X(:,k) = expm(A * t(k)) * x0;
end
```

Plot for Continuous Time $X(t)$

```
figure; hold on; grid on;
plot(t, X(1,:), 'LineWidth', 2.5);
plot(t, X(2,:), 'LineWidth', 2.5);
xlabel('time $t$', 'Interpreter','latex');
ylabel('Components of $X(t)$', 'Interpreter','latex');
title('\textbf{Assignment 6, Problem 6}', 'Interpreter','latex');
subtitle('($h=0.5$)','Interpreter','latex');
```

Approx. Discretization ($h = 0.5$)

$$A_d^k = (I + Ah)^k$$

```
h = 0.5;
k = t/h;
Ad = eye(2) + h*A
```

```
Ad = 2x2
    1.0000    0.5000
    0.5000    0.5000
```

```
%Xapp = (Ad.^k)*x0
Xapp = zeros(2, numel(k));
for kk = 1:numel(k)
    Xapp(:,kk) = (Ad^k(kk)) * x0;
end

hold on; plot(t,Xapp(1,:),'--','LineWidth', 2); plot(t,Xapp(2,:),'--','LineWidth',2);
```

Exact Discretization ($h = 0.5$)

$$A_d = e^{Ah}$$

```
h = 0.5;
k = t/h;
Ad = expm(A*h);

Xexact = zeros(2, numel(k));
for kk = 1:numel(k)
    Xexact(:,kk) = (Ad^k(kk)) * x0;
```

```
end

hold on; plot(t,Xexact(1,:),'--','LineWidth', 2); plot(t,Xexact(2,:),'--','LineWidth', 2);
hold off;
legend({'$X_1(t)$, \textit{Continuous}', ...
        '$X_2(t)$, \textit{Continuous}', ...
    '$X_1(t)$, \textit{Approx. Discretization}', ...
    '$X_2(t)$, \textit{Approx. Discretization}', ...
    '$X_1(t)$, \textit{Exact Discretization}', ...
    '$X_2(t)$, \textit{Exact Discretization}'}, ...
    'Interpreter','latex', 'Location','southoutside');
```

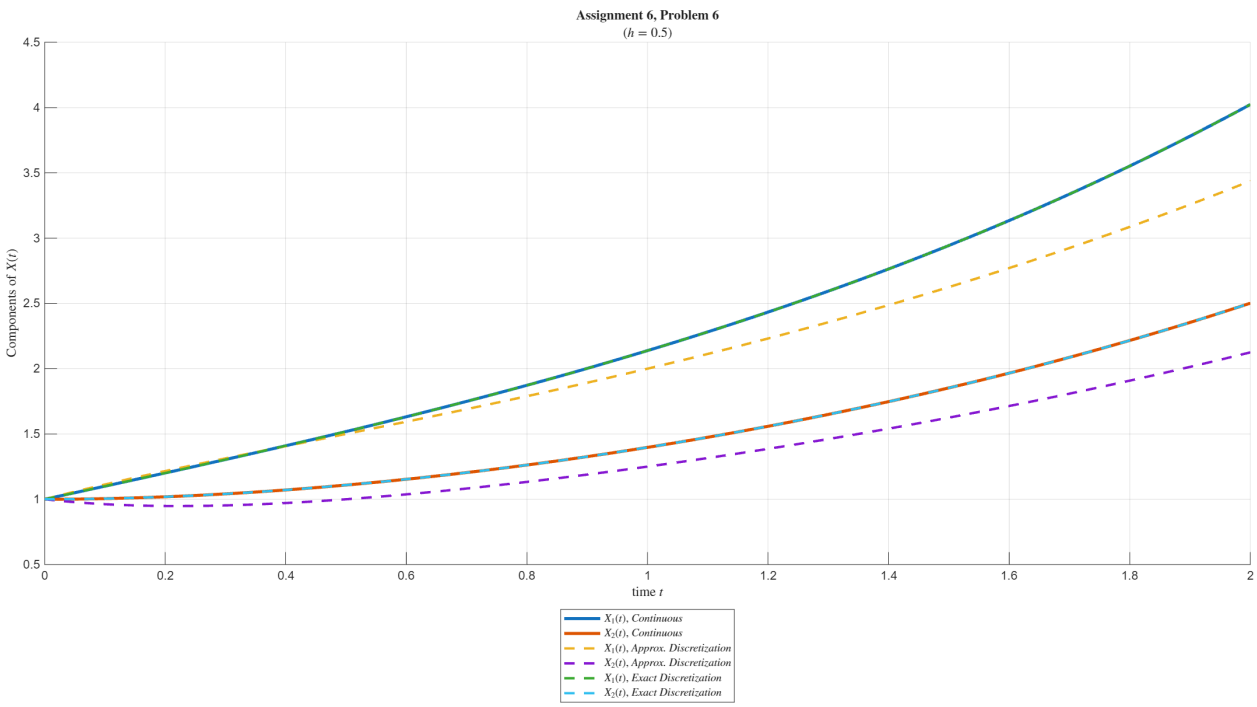

Figure 1: Continuous vs. Approximate and Exact Discretizations for a step size $h = 0.5$

```
clc;clear;close all;

% assume initial conditions since not provided
x0 = [1;1];
A = [0 1; 1 -1]
```

```
A = 2x2
     0     1
     1    -1
```

```
%[V,D] = eig(A);
t = linspace(0,2,100);
expm(A)
```

```
ans = 2x2
    1.3973    0.7410
    0.7410    0.6563
```

```
%V*expm(D)*V';
X1 =@(t) expm(A.*t)
```

```
X1 = function_handle with value:
    @(t)expm(A.*t)
```

```
X1(2)*x0
```

```
ans = 2x1
    4.0233
    2.5016
```

```

% for loop to populate X(t)
X = zeros(2, numel(t));
for k = 1:numel(t)
    X(:,k) = expm(A * t(k)) * x0;
end
```

Plot for Continuous Time $X(t)$

```
subplot(1,2,1); hold on; grid on;
plot(t, X(1,:), 'LineWidth', 3);
plot(t, X(2,:), 'LineWidth', 3);
xlabel('time $t$', 'Interpreter','latex');
ylabel('Components of $X(t)$', 'Interpreter','latex');
title('\textbf{Assignment 6, Problem 6}', 'Interpreter','latex');
subtitle('\textit{Continuous vs. Approximate Discretization } ($h=0.005$)','Interpreter','
    latex');
```

Approx. Discretization ($h = 0.005$)

$$A_d^k = (I + Ah)^k$$

```
h = 0.005;
k = t/h;
Ad = eye(2) + h*A
```

```
Ad = 2x2
    1.0000    0.0050
    0.0050    0.9950
```

```
%Xapp = (Ad.^k)*x0
Xapp = zeros(2, numel(k));
for kk = 1:numel(k)
    Xapp(:,kk) = (Ad^k(kk)) * x0;
end

hold on; plot(t,Xapp(1,:),'--','LineWidth', 3); plot(t,Xapp(2,:),'--','LineWidth',3);
hold off;
legend({'$X_1(t)$, \textit{Continuous}', ...
        '$X_2(t)$, \textit{Continuous}', ...
        '$X_1(t)$, \textit{Approx. Discretization}', ...
        '$X_2(t)$, \textit{Approx. Discretization}'}, ...
        'Interpreter','latex', 'Location','southoutside');
```

Exact Discretization ($h = 0.005$)

$$A_d = e^{Ah}$$

```
h = 0.005;
k = t/h;
Ad = expm(A*h);

Xexact = zeros(2, numel(k));
for kk = 1:numel(k)
    Xexact(:,kk) = (Ad^k(kk)) * x0;
end

subplot(1,2,2); hold on; grid on;
plot(t, X(1,:), 'LineWidth', 3);
plot(t, X(2,:), 'LineWidth', 3);
xlabel('time $t$', 'Interpreter','latex');
ylabel('Components of $X(t)$', 'Interpreter','latex');
title('\textbf{Assignment 6, Problem 6}', 'Interpreter','latex');
subtitle('\textit{Continuous vs. Exact Discretization } ($h=0.005$)','Interpreter','latex');
hold on; plot(t,Xexact(1,:),'--','LineWidth', 3,'Color','#32CD32'); plot(t,Xexact(2,:),'--',
    'LineWidth', 3,'Color','#ADD8E6');
hold off;
legend({'$X_1(t)$, \textit{Continuous}', ...
        '$X_2(t)$, \textit{Continuous}', ...
        '$X_1(t)$, \textit{Exact Discretization}', ...
        '$X_2(t)$, \textit{Exact Discretization}'}, ...
        'Interpreter','latex', 'Location','southoutside');
```

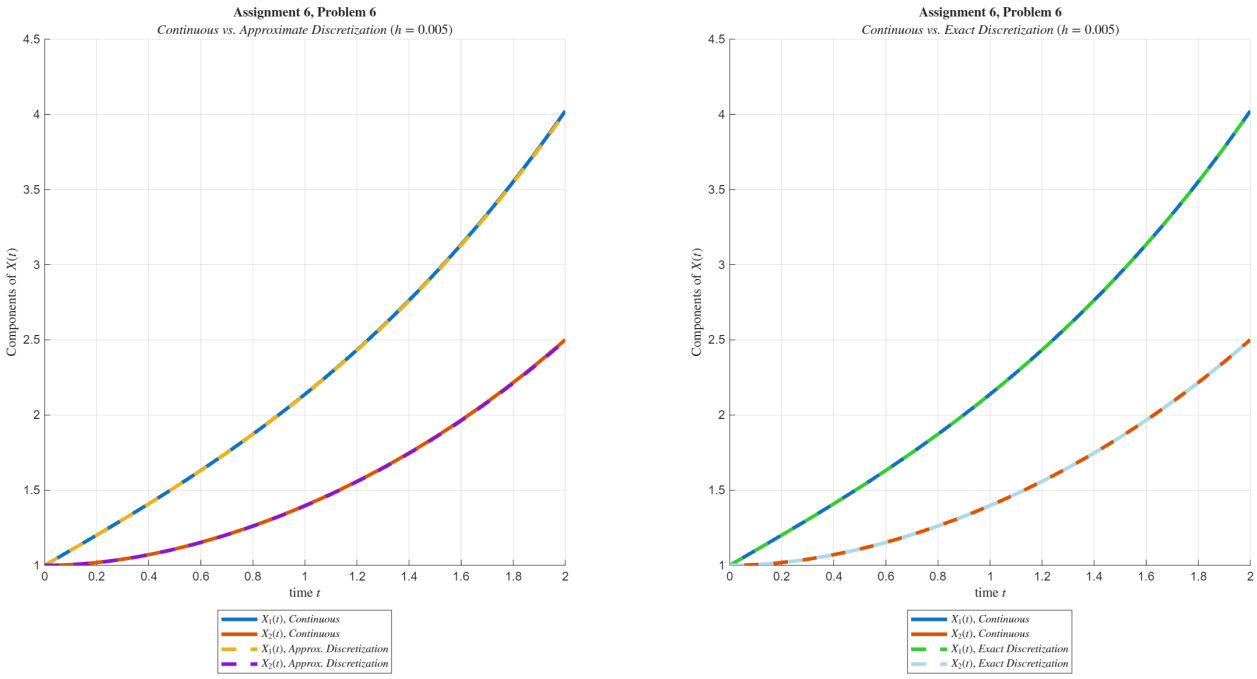

Figure 2: Continuous vs. Approximate and Exact Discretizations for a much smaller step size, $h = 0.005$

(v) Comment on the quality of the approximations from these plots. Make sure to comment on the two following quantities:

(1) For a fixed h, which approximation is closer is the solution of the continuous time system

As in Fig. 1, the exact discretization results in a much better result than the approximate discretization. For a smaller step size, the approximate $X_1(t)$ component (yellow dashed line) deviated significantly from the continuous time reference.

(2) As h decreases, do the two discretization methods (exact and approximate) become indistinguishable?

As shown in Fig. 2, both the approximate and exact discretizations become indistinguishable as h becomes sufficiently small.